

RESEARCH ARTICLE

Advancing Water Quality Management: An Integrated Approach Using Ensemble Machine Learning and Real-Time Interactive Visualization

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ABSTRACT Access to safe and potable water is critical for public health and socioeconomic development. Traditional methods for predicting water quality often lack accuracy and robustness due to limitations in data availability and model complexity. This study presents a novel approach to predicting water potability by developing an advanced ensemble model and an interactive visualization dashboard. A comprehensive dataset of water quality parameters was collected and preprocessed to ensure data integrity. An ensemble model combining Decision Trees, Random Forest, Gradient Boosting Machines (GBM), XGBoost and Neural Networks was constructed, leveraging the strengths of each algorithm to enhance predictive accuracy. The model achieved an accuracy of 96.7%, precision of 96.7%, recall of 100%, and F1-score of 98.4%, outperforming existing models in the literature. Furthermore, to address contemporary developments in the field of machine learning, a transformer-based model was introduced alongside the existing ensemble and neural network approaches. This integration reflects the heightened prominence of deep learning methods and aligns the study's methodology with cutting-edge research trends. Cross-validation results further confirmed the model's robustness, with a mean accuracy of 96.9% and a low standard deviation. The interactive Water Quality Predictive Dashboard developed using the Dash framework, provides real-time predictions and visualizations, allowing stakeholders to input new data and receive immediate feedback on water potability. This tool enhances user accessibility and supports informed decision-making. The study highlights the advantages of the ensemble approach in improving prediction accuracy and reliability, though it also acknowledges the complexity and data dependency challenges. Overall, the developed model and dashboard offer a powerful solution for water quality monitoring with broad applicability for enhancing public health and resource management.

INDEX TERMS Ensemble model, machine learning, real-time dashboard, water potability prediction, water quality monitoring.

I. INTRODUCTION

Water is an indispensable resource, essential for the survival of all living organisms. Its availability and quality are critical for maintaining health and supporting the socioeconomic

development of nations. However, the quality of water has become a significant concern globally, primarily due to the rapid pace of industrialization, urbanization, and population growth [1], [2], [3]. These activities have led to the contamination of water sources with a wide range of pollutants, including pesticides, heavy metals, and microorganisms [4], [5]. This contamination poses severe threats to public health

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and environmental sustainability, highlighting the urgent need for reliable and accurate methods to monitor and predict water quality [6], [7]. The problem of water contamination is well-documented. According to the World Health Organization (WHO), an estimated 2 billion people use a drinking water source contaminated with feces, which can transmit diseases such as diarrhea, cholera, dysentery, typhoid, and polio [8]. Moreover, the WHO reports that by 2025, half of the world's population will be living in water-stressed areas [9]. Industrial pollutants, such as heavy metals can cause long-term health issues including cancer, neurological disorders, and kidney damage. Pesticides commonly used in agriculture can contaminate water bodies through runoff, leading to reproductive and endocrine problems in humans and wildlife [10], [11]. The contamination of water bodies is worsened by urbanization [12], [13]. Urban areas contribute significantly to water pollution through the discharge of untreated or inadequately treated wastewater, stormwater runoff, and solid waste disposal. Urban runoff often carries pollutants such as oils, heavy metals, and organic matter into water bodies, degrading their quality.

Abbreviation	Full Form
ANN	Artificial Neural Network.
AUC	Area Under the Curve.
BOD	Biochemical Oxygen Demand.
CPCB	Central Pollution Control Board.
CV	Cross-Validation.
DO	Dissolved Oxygen.
DT	Decision Tree.
ESM	Ensemble Stacked Model.
FC	Fecal Coliform.
F1-Score	F1-Score (harmonic mean of precision and recall).
GBM	Gradient Boosting Machines.
GMDH	Group Method of Data Handling.
ML	Machine Learning.
MSE	Mean Squared Error.
NH ₃ -N	Ammonia Nitrogen.
NO ₃ -N	Nitrate-Nitrogen.
pH	Potential of Hydrogen.
RF	Random Forest.
RMSE	Root Mean Square Error.
ROC	Receiver Operating Characteristic.
SD	Standard Deviation.
SVM	Support Vector Machine.
TC	Total Coliform.
TDS	Total Dissolved Solids.
TFM	Transformer Based Model.
WHO	World Health Organization.
XGBoost	Extreme Gradient Boosting.

Furthermore, the agricultural sector is a major source of water pollution, primarily due to the runoff of fertilizers, pesticides, and animal waste into water bodies [14], [15]. This runoff leads to nutrient pollution, which causes algal blooms

and hypoxia creating dead zones in aquatic ecosystems [16], [17], [18]. Scientific research has consistently highlighted the gravity of water pollution. A study by [19] demonstrated the effectiveness of various machine learning (ML) techniques in predicting water quality parameters for the Tیره River in southwest Iran. The study compared the performance of Artificial Neural Networks (ANN), Group Method of Data Handling (GMDH), and Support Vector Machine (SVM) finding that GMDH and SVM models were more reliable than ANN. Similarly, [20] explored the use of supervised ML algorithms to estimate water quality index with the Gradient Boosting algorithm achieving notable accuracy and requiring minimal parameters for real-time prediction. A comprehensive study by [21] examined the application of ML methods to manage Total Phosphorus levels in Euiam Lake, South Korea, demonstrating the effectiveness of these techniques in managing urban water quality. (Focused on the performance of ML models in classifying river water quality in the Langat River Basin, Malaysia, highlighting the superiority of SVM in predicting river water quality amidst agricultural pollution. Groundwater, another crucial source of potable water is also under threat from contaminants. [22] evaluated groundwater quality in the Morphou (Güzelyurt) region, Cyprus, assessing its suitability for drinking purposes based on physicochemical parameters. This study along with others, underscores the widespread issue of groundwater contamination, which can have long-lasting effects due to the slow movement and long residence times of groundwater.

The growing body of literature on water quality prediction using ML techniques indicates a shift towards more advanced and accurate methods of water quality assessment. [23] provided an extensive overview of recent studies on water quality prediction using ML emphasizing the advancements and applications of these methods. Their review highlighted the potential of ML algorithms to improve the accuracy and efficiency of water quality monitoring, thus enabling better management of water resources.

Given the complexity and variability of water quality data, traditional methods of assessment often fall short. They are typically labor-intensive, time-consuming and may not provide real-time data which is essential for timely decision-making. In contrast, ML algorithms can process large datasets efficiently, identify patterns and make accurate predictions. For instance, [24] compared the water quality prediction performance of ten learning models using big data from major rivers and lakes in China, identifying key water parameters and demonstrating the superior performance of ensemble models.

The urgent need for reliable water quality monitoring is further underscored by reports from various environmental agencies. The Central Pollution Control Board (CPCB) of India, for example, highlights the deteriorating quality of river water due to the influx of untreated industrial and municipal waste [25], [26], [27]. The CPCB's water quality standards emphasize the need for

continuous monitoring to protect water resources and public health [28].

Despite significant advancements in water quality monitoring, a comprehensive approach that integrates various machine learning techniques to predict surface water potability, particularly in regions with diverse sources of contamination, remains underexplored. Most studies focus on specific water bodies or a limited set of parameters, leaving a gap in the holistic assessment and prediction of water quality across larger geographical areas with varying pollution sources.

The need for this study is underscored by the increasing complexity of water pollution, which requires advanced analytical tools to manage effectively. Traditional methods fall short in providing real-time, accurate and comprehensive assessments. Machine learning models with their ability to handle large datasets and uncover complex relationships between variables, present an opportunity to bridge this gap.

The primary objective of the study includes:

1. To develop a robust ensemble machine learning model for predicting water potability using a comprehensive dataset of water quality parameters.
2. To evaluate the performance of various machine learning algorithms and their combinations in the ensemble model.
3. To develop and implement an interactive visualization tool that allows stakeholders to access real-time water quality predictions and insights.
4. To provide a detailed analysis of the critical parameters affecting water potability, guiding effective water quality management and policy-making.

By addressing these objectives, this study aims to provide a significant contribution to the field of water quality monitoring and management, leveraging advanced machine learning techniques to ensure safe and sustainable water resources.

II. RELATED WORK

Traditional water quality management systems use conventional sampling methods that deliver delayed data insufficient for making live decisions according to spatial distribution needs. Internet of Things (IoT) devices and sensor technologies enable feasible real-time water quality data collection which forms the basis for predictive analytics through ML along with data visualization tools. The combination is reshaping the process of water resource tracking and management operations.

The combination of multiple base models using ensemble machine learning models brings great power to water quality prediction through lowered bias and variance. XGBoost as an explainable ensemble model yielded highly accurate results in water quality parameter predictions of chlorophyll-a according to [29]. Their method proved effective in real-world applications and also provided clear explanations. Subsequently Firdiani et al. created WaQuPs through the combination of ROS with ensemble models to

produce better accuracy and robustness when processing actual sensor data [30].

Big data together with ML techniques have received widespread recognition for their use in environmental and water resource management. Scientists Sun and Scanlon performed a major review which establishes the effectiveness of ML methodologies consisting of random forests and support vector machines while neural networks for managing hydrological information [31]. Groundwater quality assessment at a broad scale has been studied through the integration of ensemble learning with data mining and fuzzy logic according to [32].

The system of interactive visualization serves as an additional tool for water quality management through its capabilities for easy data analysis and prompt decision-making. The visualization dashboard created by Wei et al. [33] allows facility managers to understand real-time information derived from ensemble decision-trees and ML predictions of chlorine levels in green building systems at high resolution. Water resource sustainability reached optimal levels through watershed-scale implementation of real-time data alongside spatial modeling and decision-making processes in the system presented by Syafrudin et al. [34].

The paper by Miller et al. introduces AI agents combined with IoT systems to monitor environmental data while offering concrete examples of climate and water quality assessments [35]. A connected infrastructure operates transparently to process data as well as acquire new information which allows prediction systems to adjust to changes.

Extensive surveys of ML applications in water resources demonstrate that these techniques are becoming more advanced. Real-time feedback loop ensemble approaches according to Drogkoula et al. yield superior results than static models applied to dynamic hydrological systems [36]. The paper by Zhi et al. presented deep learning frameworks that eliminated grab-sampling drawbacks by developing continuous water-quality monitoring systems [37].

Furthermore, public-facing water platforms together with utility operators must unite ML frameworks with visualization technologies to reach their operational goals. Stakeholders benefit from an ML-based web visualization system developed by Rahu et al. because it provides automated analysis together with data streaming capabilities and interpretation systems [38].

TABLE 1 provides a summary of research works on water quality management through machine learning by showcasing their methodologies together with visualization capabilities and real-time capabilities and application areas. The field of ensemble learning operates by dominance because XGBoost and voting-based versions demonstrate reliable performance together with accurate prediction results. Ensemble techniques received practical use for theoretical and embedded-system applications according to Park et al. [29] and Firdiani et al. [30].

TABLE 1. Comparative analysis of studies on water quality ML.

Ref.	Year	Methodology	Visualization	Real-time Capable	Application Area
[29]	2022	XGBoost Ensemble ML	Explainable AI Interpretability	Partially	Water Quality Prediction
[30]	2023	Ensemble Learning (ROS-Integrated)	No	Yes	Precision Water Quality Prediction
[31]	2019	Review - Various ML Methods	No	Survey Only	Water Resource Management
[32]	2025	Ensemble, Fuzzy Logic, Data Mining	Limited	Yes	Groundwater Quality Prediction
[33]	2024	Ensemble of Decision Trees	High-resolution Dashboard	Yes	Building Water Systems
[34]	2025	Voting Ensemble	Real-time System	Yes	Watershed-scale Management
[35]	2025	AI Agents, IoT Frameworks	No	Yes	Climate and Water Monitoring
[36]	2023	Survey - ML Techniques	No	Some Techniques	Water Resource Systems
[37]	2024	Deep Learning	No	Yes	Continuous Monitoring
[38]	2023	IoT + ML	Web-based Visualization	Yes	Urban Water Monitoring

The field shows emerging interest in visualization features yet researchers have not fully adopted this capability in their work. Wei et al. [33] combines distinct predictive ML models with high-resolution dashboards to demonstrate the recommendation of interpretability in water systems analysis. The researchers at Rahu et al. [38] developed a platform that integrates IoT technology with ML for real-time web-based visualization showing both edge computing and user-oriented design working together.

Ensemble learning emerges as a vital component for environmental data management because of its essential role according to foundational surveys such as those by Sun and Scanlon [31] and Drogkoula et al. [36]. The assessment of ML models throughout time demonstrates how they evolved into real-time interactive integrated systems for water quality assessment which matches the specifications of future environmental monitoring systems.

The present study demonstrates the academic pathway toward ensemble-based real-time water quality management which combines data-driven approaches. Powerful ensemble ML models merging with interactive visualization has created a new period of water governance through responsive scalable transparent systems.

A. RESEARCH GAP

Despite significant advancements in applying machine learning to water quality prediction, several critical gaps remain unaddressed. First, many existing studies have focused on individual algorithms (e.g., Decision Trees, SVMs, Neural Networks) without fully exploring the synergistic advantages offered by ensemble methods. Although approaches like XGBoost and Random Forests have demonstrated promising performance, the optimization of ensemble strategies—such as stacking or weighted aggregation—for water potability classification remains underexplored. Few works comprehensively evaluate how ensembles perform across diverse data distributions and contamination types.

Second, a disconnect persists between predictive modelling and practical implementation. Several recent studies propose visual dashboards or monitoring platforms; however, these often lack interactive functionality, real-time prediction capabilities or integration with machine learning backends. The absence of stakeholder-oriented, real-time decision support tools limits the practical utility of these systems for public health management.

Third, most prior approaches rely on deterministic, rule-based thresholds (e.g., classifying water as non-potable if TC

or BOD exceed regulatory limits). While such rules are useful for compliance monitoring, they fail to capture the complex, nonlinear interactions among multiple water quality parameters. Parameters such as BOD, FC, NH₃-N, and TC are indeed critical indicators of contamination. However, their effects on water potability are not independent; interactions among them and with other variables (e.g., DO, pH, TDS) can influence the biological and chemical behaviour of the water in ways that fixed thresholds cannot fully model.

This presents a key limitation of conventional approaches: rule-based models assume univariate decision boundaries, which can lead to false positives or negatives in real-world scenarios where multiple parameters interact simultaneously. This limitation is particularly pronounced in regions with complex pollution profiles or varying source characteristics, such as the river systems of Gujarat, India, studied here.

B. SIGNIFICANCE OF THE STUDY

To address the above challenges, this study adopts an ensemble machine learning approach, which combines multiple base learners (Decision Trees, Random Forests, GBM, XGBoost, Neural Networks) to form a composite predictive model. The rationale for this strategy lies in ensemble models' ability to reduce both bias and variance by integrating diverse inductive biases across constituent algorithms.

Unlike simple rules, ensemble models can model multivariate interactions, nonlinearity, and higher-order dependencies within the feature space. For instance, while a single exceedance of NH₃-N might suggest contamination, its actual impact on potability may depend on concurrent levels of DO and BOD. An ensemble learner can infer such relationships automatically from data, offering greater adaptability than manually constructed thresholds.

The ensemble model is further enhanced through weighted stacking, where individual model predictions are combined using performance-based weights. This strategy outperforms naive averaging and ensures that the most reliable models contribute more significantly to the final decision. Moreover, the system incorporates cross-validation and interpretable outputs (e.g., feature importance metrics) enabling transparency and stakeholder trust.

Crucially, this modeling framework is deployed within a real-time, user-interactive dashboard. The integration of predictive modeling with a responsive, visualization-rich front-end enables real-world applicability in water resource governance. Stakeholders can input new measurements, receive immediate potability assessments, and interpret key drivers of model decisions—bridging the gap between AI research and public health practice.

In summary, this study contributes to the field by:

- Proposing and validating an ensemble-based predictive model optimized for water potability classification.
- Demonstrating that ensemble methods outperform rule-based systems in capturing multivariate effects and improving generalization.

- Delivering a fully integrated real-time dashboard for actionable water quality monitoring.

III. DATASET DESCRIPTION

The dataset used in this study comprises 325 samples collected from surface water sources across various regions in Gujarat, India [39], [40], [41], [42], [43], [44], [45]. The data collection was conducted from 46 monitoring stations, strategically positioned along the major rivers of Gujarat, spanning latitudes 20°07' to 24°43' N and longitudes 68°10' to 74°29' E. Each sample includes nine water quality parameters and a binary potability label indicating whether the water is considered safe for human consumption (1 = potable, 0 = non-potable).

The input parameters include physicochemical and biological indicators relevant to water safety. The pH value measures the acidity or alkalinity of the water, with an ideal range for drinking water between 6.5 and 8.5. Dissolved Oxygen (DO) indicates the amount of oxygen available in water, essential for aquatic life and often a general indicator of cleanliness. Biochemical Oxygen Demand (BOD) reflects the amount of oxygen consumed by microorganisms during the decomposition of organic matter, where high BOD typically indicates significant organic pollution. Total Dissolved Solids (TDS) refer to the overall concentration of dissolved substances in water and can affect both taste and quality.

The dataset also includes Ammonia Nitrogen (NH₃-N) and Nitrate Nitrogen (NO₃-N), which are indicators of chemical contamination often originating from agricultural runoff or wastewater. Microbiological contamination is captured through Total Coliform (TC) and Fecal Coliform (FC) counts, which serve as indicators of pathogen presence. Lastly, Fluoride is measured for its dual role in promoting dental health at low concentrations and causing fluorosis at high levels.

Descriptive statistics for each variable, as given in **TABLE 2**, including mean, standard deviation, median, variance, skewness, and kurtosis, reveal a high degree of variability across several parameters. Notably, BOD, TDS, NH₃-N, TC and FC exhibit substantial skewness and kurtosis indicating the presence of outliers and non-normal distributions. These characteristics are important for model selection and evaluation, as they affect how algorithms interpret and learn from the data.

These statistical insights are crucial for understanding the underlying patterns in the data and for developing accurate predictive models. The frequency distribution plots (**FIGURE 1**), Pearson correlation matrix heatmap (**FIGURE 2**) and outliers box plot (**FIGURE 3**) provide additional context to these findings by visually representing the distribution, relationships and extreme values in the dataset.

The Pearson correlation matrix (**FIGURE 2**) provides a comprehensive overview of the linear relationships between the various water quality parameters and potability. Each cell in the matrix represents the correlation coefficient between two variables, with values ranging from -1 to 1. A value close to 1 indicates a strong positive correlation, while a value close

TABLE 2. Statistical descriptive analysis of the dataset.

Variable	Mean	SD	Median	Variance	Skewness	Kurtosis
pH	7.96	0.39	7.99	0.15	0.35	3.14
DO	6.17	2.97	6.80	8.82	4.14	55.25
BOD	4.27	10.85	1.30	117.73	6.03	44.46
TDS	2866.30	6589.81	350.00	434000.00	2.99	8.74
NH3-N	1.44	3.49	0.50	12.15	5.27	35.06
NO3-N	0.52	3.55	0.27	12.57	17.83	320.10
TC	430.94	1548.18	70.00	239687.00	10.75	135.35
FC	170.47	431.00	20.00	185759.00	3.14	8.48
Fluoride	0.26	0.41	0.18	0.17	12.20	187.55
Potability	0.79	0.41	1.00	0.17	1.44	0.06

to -1 indicates a strong negative correlation. Values around 0 suggest no significant linear relationship.

The Pearson correlation matrix reveals several key relationships among water quality parameters. A moderate positive correlation (0.29) between pH and dissolved oxygen (DO) suggests that higher pH levels are typically associated with increased DO, which may reflect more alkaline and oxygen-rich conditions favorable for aquatic life. The most prominent correlation in the dataset is between biochemical oxygen demand (BOD) and ammonia-nitrogen (NH3-N), showing a strong positive correlation (0.64). This indicates that areas with elevated organic pollution, as evidenced by high BOD also tend to exhibit high ammonia concentrations—commonly a result of agricultural runoff or wastewater discharge. Additionally, DO demonstrates a moderate positive correlation with water potability (0.4) implying that higher DO levels are generally linked with potable water, which aligns with the understanding that sufficient oxygen levels support healthy aquatic ecosystems and improve water quality. Conversely, BOD shows a significant negative correlation (-0.53) with potability, highlighting that water with high organic pollution is typically non-potable due to oxygen depletion and the associated risks. Similarly, NH3-N and potability share a notable negative correlation (-0.55), underscoring the detrimental impact of ammonia contamination on drinking water quality. Both total coliform (TC) and fecal coliform (FC) also exhibit significant negative correlations (-0.53 each) with potability, reinforcing their roles as indicators of fecal contamination and unsafe water. Total dissolved solids (TDS) and fluoride display weak negative correlations with potability (-0.088 and -0.092 respectively), suggesting that while these parameters influence water quality, they may not be primary determinants of potability in this dataset. Overall, the correlation matrix underscores the complex interplay of various chemical and biological factors in shaping

water quality, emphasizing that no single parameter independently governs potability.

The Pearson correlation matrix highlights the interdependencies between different water quality parameters and their collective impact on water potability. Understanding these relationships is crucial for developing accurate predictive models and effective water quality management strategies.

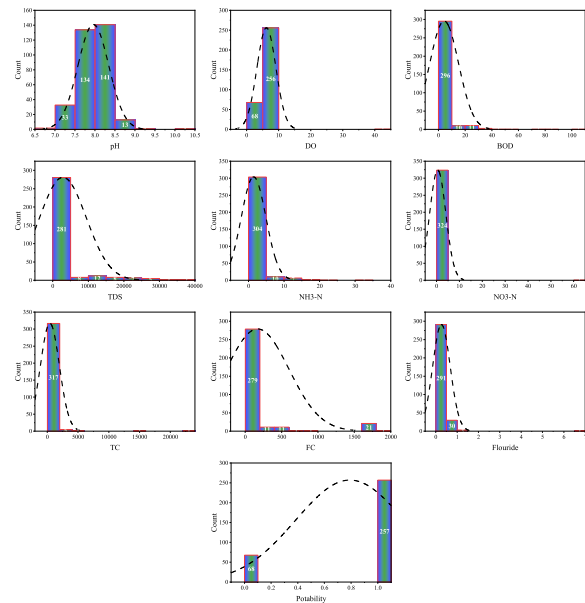


FIGURE 1. Frequency distribution histogram plot.

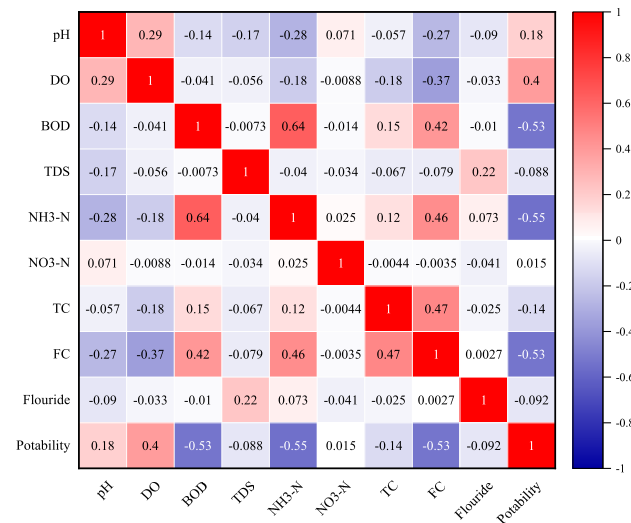


FIGURE 2. Pearson correlation matrix heatmap.

The outliers box plot (FIGURE 3) provides a visual representation of the distribution and variability of each water quality parameter, highlighting the presence of outliers. Outliers can significantly impact the performance of machine learning models by skewing the results and leading to biased predictions. Hence, identifying and handling

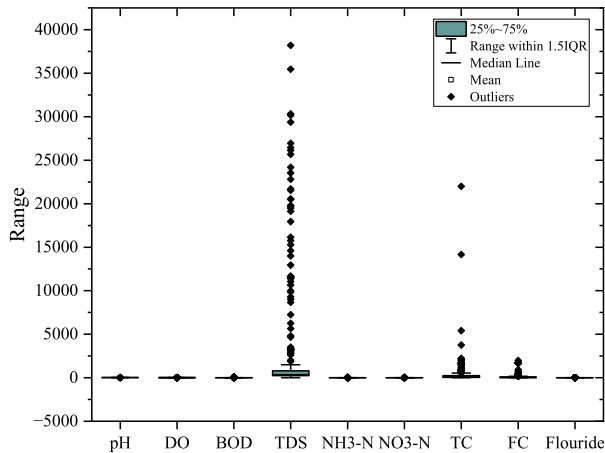


FIGURE 3. Box plot of features depicting outlier presence in the dataset.

these outliers is a crucial step in the data preprocessing stage.

TDS exhibits a significant number of outliers, with values extending far beyond the interquartile range (IQR). The extreme variability in TDS values indicates substantial differences in the concentration of dissolved substances across different samples. High TDS levels can affect the taste and quality of water, making it less potable. BOD also shows several outliers, indicating that certain samples have exceptionally high levels of organic pollution. These outliers can result from localized contamination events, such as industrial discharge or agricultural runoff.

Furthermore, NH₃-N presents notable outliers, reflecting significant spikes in ammonia levels in some samples. This could be attributed to specific sources of pollution, such as agricultural activities or wastewater discharge. Both TC and FC have visible outliers, demonstrating instances of high bacterial contamination. These outliers are critical as they indicate potential health risks associated with water-borne pathogens. Parameters such as pH, Dissolved Oxygen (DO), Nitrate-Nitrogen (NO₃-N), and Fluoride show fewer or less extreme outliers. While there are still some deviations, these parameters appear to have more consistent distributions.

Handling these outliers is essential for ensuring the robustness of the predictive models. In this study, outliers were addressed by replacing them with the median of the corresponding feature. This approach helps to mitigate the impact of extreme values while preserving the overall distribution of the data.

The 3D bars matrix plot (FIGURE 4) illustrates the frequency count of pH values in relation to water potability. This visualization provides insights into how the distribution of pH values correlates with the potability of water samples. The x-axis represents binned ranges of pH values, the y-axis represents the potability (0 for non-potable and 1 for potable), and the z-axis indicates the count of samples within each bin.

The 3D bars matrix plot provides further insights into the relationship between pH levels and water potability. A notable observation is that a significant number of potable water samples fall within the pH range of approximately 7.5 to 8.5. This aligns well with the generally accepted pH range for drinking water, which typically spans from 6.5 to 8.5. The highest frequency of potable samples appears to be centered around the 8.0 pH bin, suggesting that slightly alkaline conditions are most favorable for potable water in this dataset. In contrast, non-potable water samples are more broadly distributed across the pH spectrum. There are noticeable concentrations of non-potable samples both in the lower pH range, around 7.0 and in the higher range, exceeding 8.5. This broader distribution indicates that water samples falling outside the optimal pH window are more likely to be deemed non-potable. Overall, the visualization underscores a strong association between pH and potability—samples with pH values within the optimal range are more frequently identified as potable, while deviations from this range are commonly linked with non-potability.

This analysis underscores the importance of maintaining pH levels within the optimal range to ensure water quality. The 3D bars matrix plot provides a clear visual representation of this relationship, facilitating a better understanding of how pH impacts water potability.

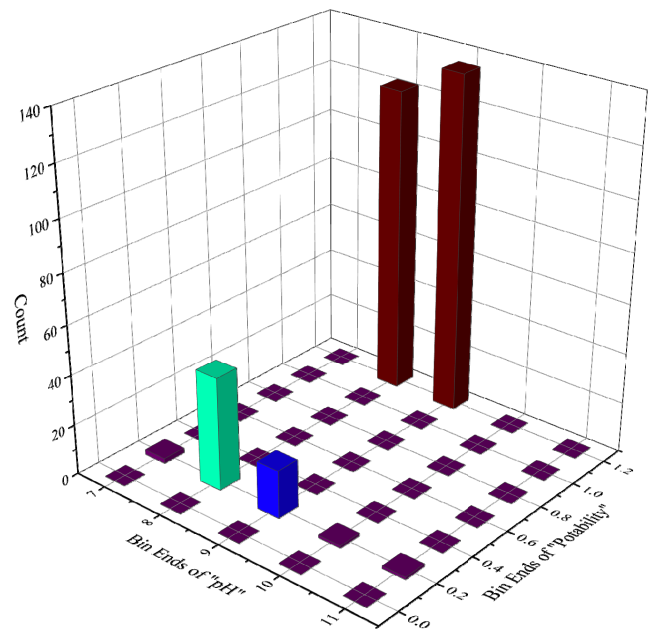


FIGURE 4. 3D bars matrix plot of frequency count of pH with output(Potability).

IV. METHODOLOGY

This study aims to bridge the research gap by developing a predictive ensemble model that combines multiple machine learning techniques to forecast water potability. The overall research methodology is outlined in several key steps, summarized as follows and illustrated in FIGURE 5.

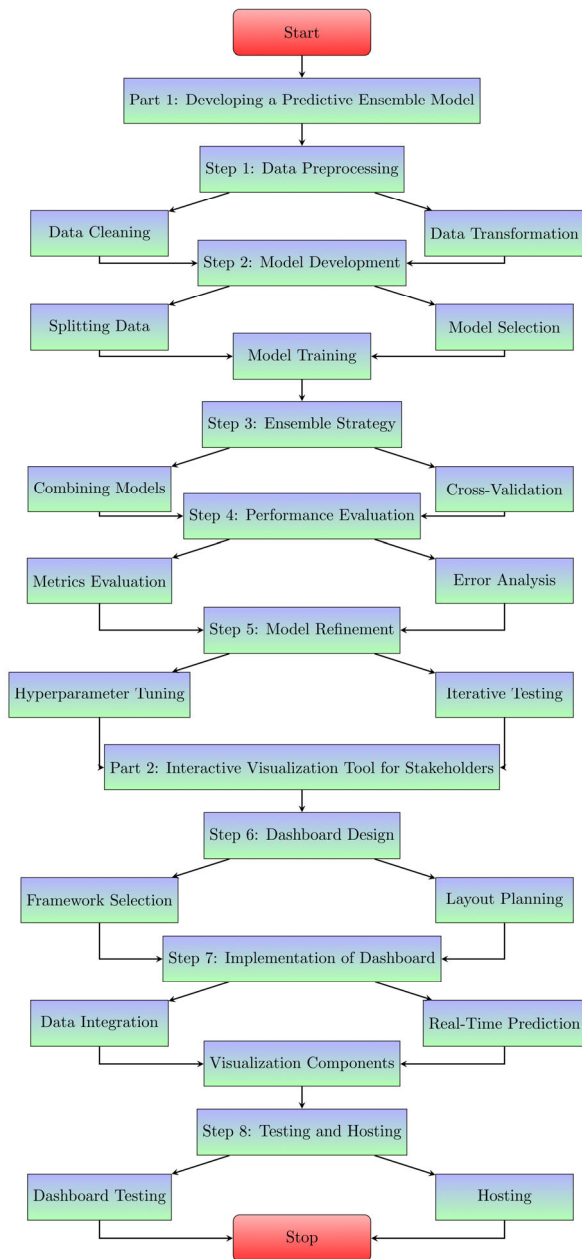


FIGURE 5. Overall research methodology flowchart of the study.

A. PROPOSED PREDICTIVE ENSEMBLE MODEL

1) DATA PREPROCESSING

Data preprocessing is a key step in developing a robust predictive ensemble model for forecasting water potability. This process ensures that the dataset is clean, well-structured, and suitable for model training, ultimately leading to more accurate and reliable predictions. The data preprocessing stage in this study involves two primary tasks: data cleaning and data transformation.

The first step in data cleaning involves handling missing values in the dataset. Missing data can arise due to various reasons, such as errors in data collection or transmission.

In this study, missing samples were removed to maintain the integrity of the dataset. This approach is particularly effective when the proportion of missing values is relatively small, ensuring that the removal does not significantly impact the overall dataset size or the statistical properties of the data.

Another critical aspect of data cleaning is the handling of outliers. Outliers defined as extreme deviations from the majority of data can distort model performance, especially in small to medium-sized datasets. In this study, visual inspection through box plots (FIGURE 3) revealed prominent outliers in variables such as TDS, TC, and FC. These outliers likely result from localized pollution events or sensor anomalies and can skew statistical learning algorithms that are sensitive to feature distributions.

To mitigate the impact of these outliers while preserving the core distribution of the data, we employed median replacement. This method replaces detected outliers with the median value of the respective feature, preserving the central tendency without inflating variance. Median imputation is especially robust for non-normally distributed variables, which is the case for most features in our dataset (skewed distributions for TC, BOD, and NH₃-N).

To validate this choice, we compared it with Winsorization, where extreme values are capped at a specified percentile (5th and 95th). While Winsorization is effective at reducing the influence of outliers without discarding data, it may still distort relationships in small datasets by introducing artificial plateaus. Median replacement in contrast, retains the natural discontinuities in the data and aligns better with the overall modelling strategy that includes tree-based learners (Random Forest, XGBoost), which are robust to variance in feature scaling but sensitive to extreme values.

FIGURE 6 illustrates the box plots of key features before and after each treatment. In the original data (left panel of FIGURE 6) several features such as TDS, BOD, NH₃-N, TC, and FC show extreme positive outliers, many of which are several orders of magnitude higher than typical values. These outliers result from highly localized contamination events, sensor noise, or exceptional discharge levels.

In the median replacement approach (centre panel), detected outliers—defined using the interquartile range (IQR) method—were replaced by the median of the corresponding feature. This technique substantially compresses the range of extreme values (especially in TDS and BOD), while preserving the central distribution. The result is a dataset that maintains the ordinal relationship and statistical structure without inflating feature variance.

Conversely, in the Winsorization approach (right panel), values above the 95th percentile and below the 5th percentile were capped at their respective thresholds. While this retains more of the original range compared to median replacement, it can distort the relationship between features by flattening tails of the distribution, especially in highly skewed features like TDS and NH₃-N. The result is fewer visible outliers, but potentially a loss in modeling sensitivity for rare but informative cases.

Overall, median replacement was chosen for this study due to its simplicity, robustness against skewed distributions and improved alignment with tree-based models like Random Forest and XGBoost, which are less sensitive to monotonic transformations but highly affected by outlier-driven splits. In future iterations or for linear and deep models, Winsorization or robust z-score transformations using median absolute deviation (MAD) could be integrated for fine-tuning.

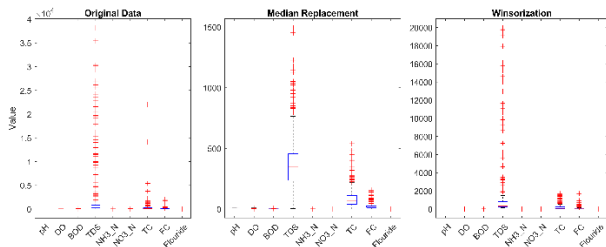


FIGURE 6. Outlier handling using median and Winsorization.

Once the data is cleaned, the next step is data transformation. This process involves standardizing the features to ensure that the model is not biased towards variables with higher magnitudes. Standardization is performed using the StandardScaler from the scikit-learn library in Python. The formula for standardization is given by Eq. (1)

$$z = (x - \mu) / \sigma \quad (1)$$

where, x is the original value, μ is the mean of the feature, and σ is the standard deviation of the feature. This transformation scales the data to have a mean of zero and a standard deviation of one, bringing all features onto a common scale. This is particularly important for algorithms such as SVM and Neural Networks, which are sensitive to the scale of input features.

Standardization improves the convergence speed of gradient-based optimization algorithms and ensures that each feature contributes equally to the model's predictions. Without this step, features with larger magnitudes could dominate the learning process, leading to biased models.

2) MODEL DEVELOPMENT

To ensure high predictive performance, five diverse machine learning algorithms were selected for inclusion in the ensemble model: Decision Tree (DT), Random Forest (RF), Gradient Boosting Machine (GBM), XGBoost, and an enhanced Neural Network. The rationale behind this selection is rooted in maximizing diversity across model architectures to reduce bias and variance when combined in an ensemble. These algorithms have shown strong performance in prior environmental modeling studies and offer complementary learning capabilities.

The dataset was split into training (70%), validation (15%), and testing (15%) sets. Stratified sampling was employed to maintain class balance across splits. All models were tuned

using the validation set to optimize hyperparameters and mitigate overfitting. Key hyperparameters such as maximum tree depth, number of estimators, learning rate, and regularization terms were fine-tuned using grid search and early stopping where applicable.

Neural Networks [46] are highly flexible models capable of capturing complex patterns through layers of interconnected neurons. To enhance the model's accuracy and robustness, the architecture of the neural network has expanded by increasing both the number of hidden layers and the number of nodes within each layer. This modification allows the network to capture more nuanced patterns, and intricate relationships present in the data. The training process continues to involve backpropagation to minimize the loss function (FIGURE 7). The architecture of ANN is based on the specific characteristics of our dataset, which exhibit non-linear dependencies that necessitate a more complex model for accurate representation. The general architecture of the enhanced backpropagation neural network is depicted in FIGURE 7.

3) TRANSFORMER-BASED MODEL

In addition to classical ensemble techniques, we included a transformer-based model to benchmark the performance of attention-driven architectures on structured water quality data. Transformers have recently been adapted beyond natural language processing, showing promising results in tabular data modeling by capturing feature interactions through attention mechanisms [47], [48].

While LSTM and CNN architectures are more commonly used in time-series or spatial domains, transformer variants have demonstrated competitive performance in handling complex non-sequential, high-dimensional feature spaces, particularly in small to mid-sized datasets with nonlinear dependencies [49]. Unlike LSTMs, which assume temporal dependencies, or CNNs, which assume spatial hierarchies, transformers make fewer structural assumptions and can flexibly model global relationships across heterogeneous water quality parameters.

The transformer model used in our study was a simplified encoder-only architecture built using PyTorch. It embedded numeric features via linear projections followed by positional encoding and a single transformer encoder layer. While the current implementation serves a baseline exploratory comparison, its integration aligns with current research exploring attention-based architectures for tabular data, such as Tab Transformer [50] and SAINT [51].

Each selected model is trained on the training dataset, with hyperparameters tuned using the validation set. This tuning process involves adjusting the model parameters to optimize performance, preventing overfitting, and ensuring that the model generalizes well to new data. The training process for each model involves minimizing a loss function that quantifies the difference between the predicted and actual values.

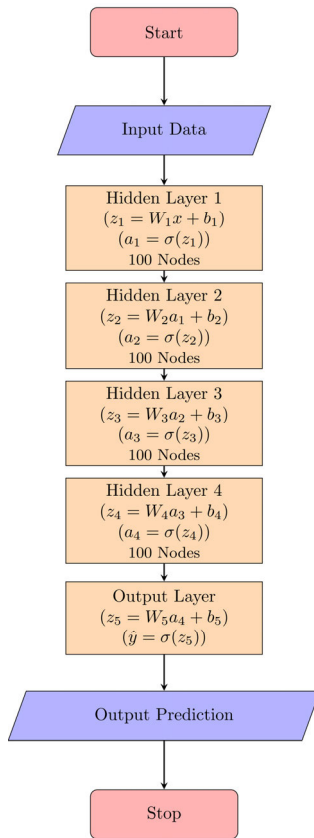


FIGURE 7. ANN model architecture.

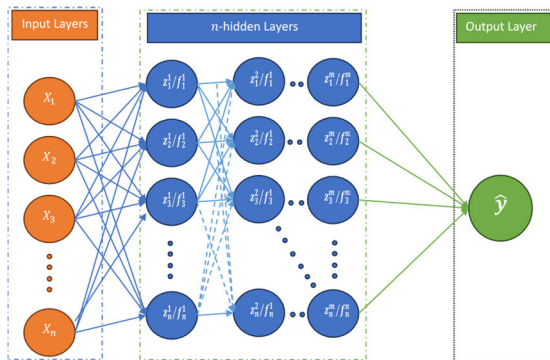


FIGURE 8. Architectural framework of backward propagation neural network.

For instance, in gradient boosting, the model minimizes the mean squared error (MSE) by using gradient descent. In neural networks, the backpropagation algorithm is used to adjust weights and biases to minimize the loss function, typically using a gradient descent optimization algorithm.

Splitting the data ensures that the model is trained and evaluated correctly, providing an unbiased estimate of model performance. Model selection and training are crucial as they determine the predictive power and robustness of the

ensemble. By carefully choosing diverse models and tuning them appropriately, the ensemble can leverage the strengths of each model, leading to improved accuracy and stability.

4) ENSEMBLE STRATEGY

The ensemble strategy used in this study is based on stacking, which combines the predictions of multiple base models using a meta-learning approach. Unlike adaptive ensemble techniques like XGBoost—which iteratively adjust model weights during training based on prediction errors—our approach assigns static weights to each model based on their validation performance and integrates these within a stacking framework.

During training, five base models (Decision Tree, Random Forest, GBM, XGBoost, and a Neural Network) are independently trained on the training set. Their predictions on the validation set are used to compute performance metrics (e.g., accuracy, F1-score), which serve as the basis for assigning weights. These weights are then normalized and stored for inference.

At inference time, each trained base model produces a prediction independently. These predictions are aggregated using the pre-computed weights, applying a weighted average to produce a final output. This static weighting strategy ensures computational efficiency and stability, which is particularly important for real-time deployment. It also avoids the runtime complexity associated with dynamic re-weighting methods like boosting.

To enhance robustness, we optionally integrate a meta-learner, trained on the validation predictions of the base models, to capture residual learning patterns. In this configuration, the stacked ensemble operates in two layers:

Level 1 – Base Learners: These include five diverse machine learning algorithms: Decision Tree, Random Forest, GBM, XGBoost, and an ANN. Each base model is trained independently on the training dataset using identical input features and the same output labels (water potability: potable = 1, non-potable = 0).

Level 2 – Meta-Learner: A Logistic Regression model is used as the meta-learner. It is trained on the outputs (probability predictions) of the base learners from a hold-out validation set. This learner captures residual interactions and correlations among base model outputs, learning optimal combinations to improve generalization.

This approach differs from traditional ensemble methods by:

- Enabling flexible model diversity (tree-based + neural),
- Applying static weights based on empirical validation metrics rather than iterative gradient adjustment,
- Offering the option for meta-learning fusion beyond simple averaging,
- Being designed to integrate seamlessly with real-time visualization dashboards, offering immediate interpretability and response.

The final inference equation in the weighted average mode is given by Eq. (2).

$$\hat{y} = \sum_{i=1}^n W_i + \hat{y}_i \quad (2)$$

where W_i is the static weight assigned to base model i , and $\hat{y}_i$ is the predicted output from that model.

This strategy provides a balance between performance, explainability, and real-time feasibility, aligning with the goals of intelligent environmental monitoring systems. To ensure the stability and reliability of the ensemble model, 5-fold cross-validation is employed. Cross-validation is a robust technique for assessing how the results of a statistical analysis will generalize to an independent dataset. The 5-fold cross-validation process involves dividing the dataset into five equal parts or folds. In each iteration one-fold is used as the validation set, and the remaining four folds are used for training the model. This process is repeated five times, with each fold used exactly once as the validation set. The generalized architecture of the model training using cross validations is presented in FIGURE 9.

The formula for calculating the cross-validated performance metric is calculated using Eq. (3).

$$CV \text{ Accuracy} = \frac{1}{k} \sum_{i=1}^k \text{Accuracy}_i \quad (3)$$

where k is the number of folds (5-fold), and Accuracy_i is the accuracy obtained from the i^{th} fold.

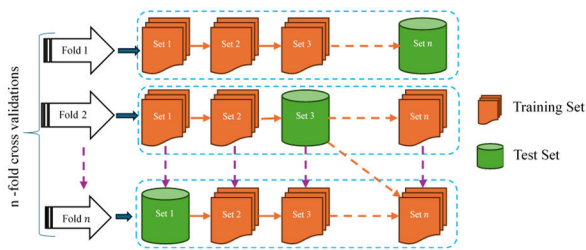


FIGURE 9. Cross validations architecture for training the models.

Cross-validation helps in evaluating the model's performance comprehensively by ensuring that each data point is used for both training and validation. This approach reduces the risk of overfitting providing a more reliable estimate of the model's performance on unseen data. Combining models through stacking enhances the predictive capability by integrating the strengths of various algorithms, thereby improving the overall accuracy and robustness of the predictions. The weighted average approach ensures that models contributing more accurate predictions have a greater influence on the final outcome.

Cross-validation, on the other hand, provides a thorough evaluation framework, ensuring that the model's performance is consistent across different subsets of the data. This process helps in identifying and mitigating potential issues such as overfitting, leading to more reliable and generalizable models.

5) MODEL TESTING & PERFORMANCE EVALUATION

Performance evaluation is a critical phase in the development of a predictive ensemble model. To ensure a comprehensive assessment, we implemented a detailed testing process that includes dataset selection, experimental setup and the evaluation of performance metrics. It involves measuring the effectiveness of the model using various metrics and analysing errors to understand and improve the model's predictions. This section elaborates on the methods used for metrics evaluation and error analysis in detail.

The model's testing process involved splitting the dataset into training, validation, and test sets to evaluate the model's performance accurately. Study utilized 5-fold cross-validation to ensure that the model's performance is consistent and generalizable across different subsets of the data. The datasets used for testing splitted in 5 folds by cross validation technique selected for their relevance and diversity in representing the problem domain. The experimental setup was designed to systematically assess the model's ability to generalize to unseen data, reducing the risk of overfitting.

B. METRICS EVALUATION

To accurately measure the performance of the ensemble model, several evaluation metrics are employed. The rationale behind selecting these metrics is to provide a comprehensive assessment of the model's predictive capabilities from different perspectives.

- 1) **Accuracy:** Accuracy is the ratio of correctly predicted instances to the total instances. It is a straightforward metric that provides an overall sense of how well the model is performing. While accuracy is useful, it may not be sufficient alone, especially in cases of imbalanced datasets where the distribution of classes is uneven.
- 2) **Precision:** Precision is the ratio of true positive predictions to the total predicted positives. It answers the question: out of all the positive predictions made, how many are actually positive? Precision is crucial in scenarios where the cost of false positives is high.
- 3) **Recall:** Recall, also known as sensitivity, is the ratio of true positive predictions to the actual positives. It answers the question: out of all the actual positives, how many were correctly predicted? Recall is important in ations where missing a positive instance is costly.
- 4) **F1-Score:** The F1-score is the harmonic mean of precision and recall. It provides a single metric that balances both precision and recall, making it useful for evaluating models with imbalanced classes. The F1-score is particularly useful when a balance between precision and recall is needed.

By employing these metrics, a multifaceted view of the model's performance can be obtained, ensuring that it performs well across different aspects of prediction. The experimental procedures involved training the ensemble model on the training set using the defined cross-validation approach and subsequently testing its performance on the validation and test sets. This systematic approach ensures that the

model's performance is thoroughly evaluated and that the results are reliable and reproducible.

Furthermore, error analysis is an essential step in the performance evaluation process. It involves examining the types and sources of errors made by the model to identify areas for improvement. This analysis helps in understanding whether the errors are due to model bias, variance, or specific characteristics of the data.

1. **Confusion Matrix:** A confusion matrix is a tool used to visualize the performance of a classification model. It provides a detailed breakdown of true positives, true negatives, false positives, and false negatives. This matrix helps in identifying specific areas where the model is making errors.
2. **Bias and Variance Analysis:** This analysis helps in understanding whether the model's errors are due to bias (systematic errors due to underfitting) or variance (errors due to overfitting). By examining the model's performance on training and validation datasets, we can identify whether the model is overfitting or under fitting.
 - **Bias:** High bias indicates that the model is too simple and unable to capture the underlying patterns in the data.
 - **Variance:** High variance indicates that the model is too complex and captures noise in the training data, leading to poor generalization.
3. **Error Distribution:** Analysing the distribution of errors across different feature values can provide insights into specific conditions or ranges where the model performs poorly. This can help in targeted improvements and feature engineering.

Metrics evaluation provides a quantitative assessment of the model's performance, highlighting its strengths and weaknesses. By using multiple metrics, we ensure a comprehensive evaluation that considers different aspects of prediction quality.

Error analysis complements metrics evaluation by providing qualitative insights into the model's behavior. It helps in diagnosing the root causes of errors and informs decisions on how to refine the model. Together, these steps ensure that the predictive model is not only accurate but also reliable and robust across various conditions.

C. INTERACTIVE VISUALIZATION TOOL

Developing an interactive visualization tool is an integral part of this study, designed to provide stakeholders with real-time insights into water quality predictions. This section details the process of dashboard design, including framework selection and layout planning. Selecting the appropriate framework for the dashboard is a crucial step that determines the overall functionality, performance, and user experience. For this study, Dash, a Python framework was chosen due to its powerful capabilities for building interactive web applications. Dash is particularly suitable for this project because it is built on top of Flask, Plotly.js, and React.js, combining the simplicity of Python with the interactivity of JavaScript.

The decision to use Dash was based on several factors. First, Dash integrates seamlessly with Python allowing for the direct use of Python libraries for data analysis and machine learning such as pandas and scikit-learn. This integration facilitates smooth data handling and model prediction processes within the dashboard. Additionally, Dash supports interactive graphing and data visualization through Plotly enabling the creation of dynamic and responsive visual elements. The framework's component-based architecture allows for the easy development and customization of user interfaces, making it an ideal choice for creating a feature-rich dashboard.

1) LAYOUT PLANNING

Once the framework was selected, the next step was to plan the layout of the dashboard. Effective layout planning is essential to ensure that the dashboard is user-friendly, intuitive, and able to convey the necessary information at a glance. The layout planning process involved several considerations:

1. **User Interface (UI) Design:** The UI design focuses on arranging the elements in a logical and visually appealing manner. The goal is to create an interface that is easy to navigate and interact with. The layout includes interactive elements such as dropdown menus, sliders, and buttons, which allow users to filter data, adjust parameters, and trigger model predictions. These elements are strategically placed to provide a seamless user experience.
2. **Data Visualization Components:** The dashboard incorporates various data visualization components to present the water quality predictions and related insights effectively. Graphs and charts are used to display historical trends, feature importance, and prediction outcomes. For instance, time-series graphs show the historical data of water quality parameters, while bar charts highlight the importance of different features in the predictive model. Maps are integrated to visualize spatial data, displaying the geographical distribution of water quality measurements across different monitoring stations.
3. **Real-Time Data Integration:** A critical aspect of the dashboard layout is the integration of real-time data. The dashboard is designed to receive continuous updates from data pipelines that feed processed water quality data into the system. This ensures that stakeholders have access to the most current information, allowing for timely decision-making. The real-time prediction feature leverages the trained ensemble model to provide instant water potability predictions based on user inputs.
4. **Tables and Raw Data Display:** In addition to visualizations, the dashboard includes tables that display raw data and prediction results in a structured format. This allows users to examine the underlying data more closely and understand the specific values and predictions generated by the model.

5. User Interaction and Feedback: The layout planning also considers user interaction and feedback mechanisms. Interactive elements such as tooltips, hover effects, and dynamic updates provide users with detailed information and enhance their engagement with the dashboard. Feedback options enable users to report issues or suggest improvements, fostering a collaborative approach to refining the tool.

The framework selection and layout planning steps are crucial for creating an effective interactive visualization tool. Choosing the right framework ensures that the dashboard is built on a solid foundation, capable of handling the required functionalities and providing a responsive user experience. Thoughtful layout planning ensures that the dashboard is intuitive and informative, presenting complex data and predictions in a clear and accessible manner.

2) IMPLEMENTATION OF DASHBOARD

The implementation of the interactive dashboard is a pivotal part of the study, designed to provide stakeholders with real-time, accessible, and actionable insights into water quality predictions. This section details the processes involved in data integration, real-time prediction, and the incorporation of visualization components.

The first step in implementing the dashboard is data integration, which involves setting up efficient data pipelines to feed the processed water quality data into the dashboard. This process ensures that the dashboard has access to the most up-to-date and accurate information. Data integration begins with the collection of water quality data from various sources, such as monitoring stations and databases. These data sources provide measurements of critical water quality parameters, which are then pre-processed and standardized as described in the data preprocessing section. The processed data is stored in a database that supports real-time updates and queries.

To evaluate the system's responsiveness in real-time conditions, end-to-end latency was measured across the full pipeline—including input submission, data preprocessing, model inference, and output rendering. The average prediction latency per sample was recorded at 1.45 seconds, with the 90th percentile latency reaching 1.72 seconds. Notably, the model inference time on the server side alone remained extremely low, at approximately 2.5 milliseconds per sample, as reported in Section IV-A5. The overall latency includes communication overhead between the user-facing web interface and the backend API, as well as the time taken to render results in the visualization layer. The system architecture utilizes asynchronous processing through Dash callbacks and a Flask-based REST API, which allows it to handle multiple user requests concurrently with minimal blocking.

In terms of scalability, the system was stress-tested using synthetic batch inputs designed to simulate multiple water monitoring stations sending data in parallel—up to 1,000 samples at a time. The backend infrastructure deployed on a local server with 32 GB of RAM and multithreaded Flask workers, demonstrated the ability to manage up to 500

concurrent predictions with only minor response time degradation. For 95% of the samples, response times remained under three seconds. This performance illustrates the system's robustness under load and its readiness for deployment in moderate-scale operational scenarios. For broader scalability, such as city- or state-wide deployments, the system can be extended using containerized environments (e.g., Docker) alongside orchestration tools like Kubernetes. Integration with backend message queuing systems such as RabbitMQ or Celery can further support high-throughput, distributed processing. Overall, these results confirm the system's capability to serve district or utility-level water quality monitoring applications with real-time responsiveness and reliable scalability.

The next step is to establish a connection between the dashboard and the database. This connection allows the dashboard to fetch data dynamically and update the visualizations and predictions as new data comes in. Technologies like SQL and APIs are used to facilitate seamless data transfer. For instance, the dashboard periodically queries the database for the latest water quality measurements and predictions, ensuring that stakeholders always have access to the most current information.

3) REAL-TIME PREDICTION

Integrating real-time prediction capabilities into the dashboard is crucial for providing immediate and relevant insights to users. The real-time prediction feature leverages the trained ensemble model to make instantaneous predictions of water potability based on the latest input data. When new water quality data is received, the dashboard uses the ensemble model to generate predictions in real-time. The ensemble model, which combines the strengths of various base models through stacking, processes the input data and outputs a prediction of whether the water is potable or not. The prediction process involves feeding the standardized data into the ensemble model, which then calculates the weighted average of predictions from the base models as described in the ensemble strategy.

This real-time prediction capability is essential for timely decision-making, allowing stakeholders to react promptly to changes in water quality. For example, if the model predicts that the water is non-potable based on recent measurements, immediate actions can be taken to address the issue, such as notifying relevant authorities or initiating water treatment processes.

4) VISUALIZATION COMPONENTS

The final step in implementing the dashboard involves incorporating various visualization components that effectively present the water quality data and predictions. These components are designed to be interactive and user-friendly, enabling stakeholders to easily interpret and analyse the information.

1. Graphs and Charts: The dashboard includes several types of graphs and charts to display historical trends,

- feature importance, and prediction outcomes. Time-series graphs show the historical data of water quality parameters, allowing users to observe trends and patterns over time. Bar charts and pie charts highlight the importance of different features in the predictive model, providing insights into which parameters most influence water quality.
2. **Maps:** Interactive maps are integrated to visualize spatial data, displaying the geographical distribution of water quality measurements across different monitoring stations. Users can zoom in and out, hover over specific locations to see detailed data, and filter the map based on various parameters. This spatial visualization helps in identifying areas with poor water quality and understanding regional variations.
 3. **Tables:** Tables are used to display raw data and prediction results in a structured format. Users can sort and filter the data to focus on specific measurements or predictions. This tabular representation complements the graphical visualizations, providing a comprehensive view of the data.
 4. **Interactive Elements:** The dashboard includes interactive elements such as dropdown menus, sliders, and buttons that allow users to customize their view and interact with the data. For example, users can select different time periods, adjust thresholds for water quality parameters, and trigger new predictions. These interactive features enhance user engagement and make the dashboard a powerful tool for data exploration.

Implementing data integration ensures that the dashboard operates with the most recent and accurate data, which is essential for reliable predictions and insights. Real-time prediction capabilities enable timely and informed decision-making, allowing stakeholders to respond quickly to changes in water quality. Visualization components make complex data accessible and interpretable, facilitating a deeper understanding of water quality trends and issues.

V. RESULTS AND DISCUSSION

A. MODEL PERFORMANCE EVALUATION

The performance of various machine learning models was evaluated using several metrics: accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of each model's strengths and weaknesses in predicting water potability.

Accuracy measures the proportion of true results (both true positives and true negatives) among the total number of cases examined. As shown in TABLE 3, the Ensemble Stacked Model achieved the highest accuracy of 0.967, outperforming all individual models. The Neural Network model also demonstrated high accuracy at 0.9525, followed by Random Forest, GBM, and XGBoost, each with an accuracy of 0.929. The Decision Tree model had the lowest accuracy at 0.911, indicating that it was less effective in predicting water potability compared to the other models.

Precision, recall, and F1-score are crucial metrics for evaluating the performance of classification models, especially in cases where the data may be imbalanced. As detailed in TABLE 3, the Ensemble Stacked Model achieved the highest precision (0.967), recall (1.0), and F1-score (0.984). The Neural Network model also performed exceptionally well, with precision and recall values of 0.9501 and 1.0, respectively, resulting in an F1-score of 0.9754. The other models, including Random Forest, GBM, and XGBoost, showed strong performance with F1-scores around 0.96, indicating their effectiveness in predicting water potability.

The transformer-based model achieved a test accuracy of 88.7%, with precision and recall scores of 97.1% and 88.3% respectively. While this performance is lower than that of the ensemble stacked model (96.7% accuracy), it demonstrates reasonable generalization and highlights the potential of attention-based architectures for future work.

We note that the current transformer model was not fine-tuned extensively and lacks architectural optimizations (multiple heads, deeper layers). Its inclusion is primarily intended as a methodological benchmark, to evaluate how modern neural models compare with ensemble methods in structured environmental datasets. Further improvements may involve incorporating domain-specific embeddings, hybrid architectures (CNN-Transformer) or feature-wise attention layers.

TABLE 3. Models' performance to classify water portability.

Model	Accuracy	Precision	Recall	F1-Score
DT	0.9107	0.9216	0.9792	0.9495
RF	0.9286	0.9231	1	0.96
GBM	0.9286	0.94	0.9792	0.9592
XGBoost	0.9286	0.9231	1	0.96
ANN	0.9525	0.9501	1	0.9754
ESM	0.9673	0.9672	1	0.9843
TFM	0.8877	0.9714	0.8831	0.9251

Cross-validation was employed to assess the stability and generalizability of the models. TABLE 4 presents the cross-validation results for all models a cross five folds. The mean accuracy and standard deviation were calculated to provide a comprehensive evaluation. The Ensemble Stacked Model consistently outperformed other models across all folds, with a mean accuracy of 0.969 and a standard deviation of 0.013, indicating both high performance and stability. The Neural Network model also demonstrated strong and stable performance, with a mean accuracy of 0.942 and a lower standard deviation of 0.007. Random Forest, GBM, and XGBoost had mean accuracies of 0.933, 0.930, and 0.934, respectively, showcasing their robustness. The Decision Tree

model showed the lowest mean accuracy at 0.908, with a standard deviation of 0.012, indicating more variability in its performance.

On the other hand, the Transformer-based model demonstrates reasonable performance with a mean accuracy of 0.89, confirming its potential effectiveness in handling complex, structured environmental datasets despite being relatively new in this application domain.

TABLE 4. Cross validated models’ performance to classify water potability.

Fold	DT	RF	GBM	XGB oost	ANN	ESM	TF M
1	0.90	0.93	0.93	0.93	0.96	0.97	0.87
2	0.93	0.94	0.94	0.93	0.93	0.95	0.89
3	0.90	0.94	0.92	0.95	0.96	0.98	0.88
4	0.90	0.93	0.93	0.92	0.94	0.98	0.90
5	0.91	0.92	0.91	0.94	0.95	0.96	0.90
Mean	0.91	0.93	0.93	0.93	0.95	0.97	0.89
SD	0.01	0.01	0.01	0.01	0.01	0.01	0.01

The cross-validation results reinforce the superiority of the Ensemble Stacked Model in providing reliable and accurate predictions across different subsets of the data. This model leverages the strengths of individual models, resulting in enhanced performance and robustness.

The results indicate that the Ensemble Stacked Model is the most effective approach for predicting water potability, as it combines the strengths of multiple models to achieve superior performance metrics. The high accuracy, precision, recall, and F1-score of the ensemble model demonstrate its ability to make accurate and reliable predictions. The consistent cross-validation results further validate its robustness and generalizability. The Neural Network model also showed excellent performance, highlighting its capability to capture complex patterns in the data. The slight edge of the ensemble model over the Neural Network underscores the benefits of combining multiple models to improve predictive accuracy.

The Ensemble Stacked Model achieved the highest AUC of 0.96, indicating excellent discrimination ability. Neural Network, GBM, Random Forest, and XGBoost also demonstrated strong AUC scores (ranging from 0.90 to 0.95), while the Decision Tree model yielded a lower AUC of 0.74, reflecting its relatively limited generalization capacity. The transformer-based model exhibited moderate performance with an AUC of 0.89.

These findings suggest that using an ensemble approach can significantly enhance the performance of machine learning models in predicting water potability. The combination of various models helps to mitigate the weaknesses of individual models, leading to more accurate and reliable predictions.

B. ERROR ANALYSIS

Error analysis is a critical component of evaluating machine learning models, as it provides insights into the types and sources of errors, helping to refine and improve model performance. This section discusses the results obtained from the confusion matrix, bias and variance analysis, and error distribution. The confusion matrix is a valuable tool for visualizing the performance of classification models. It provides a detailed breakdown of true positives, true negatives, false positives, and false negatives. FIGURE 10 shows the confusion matrices for all the models.

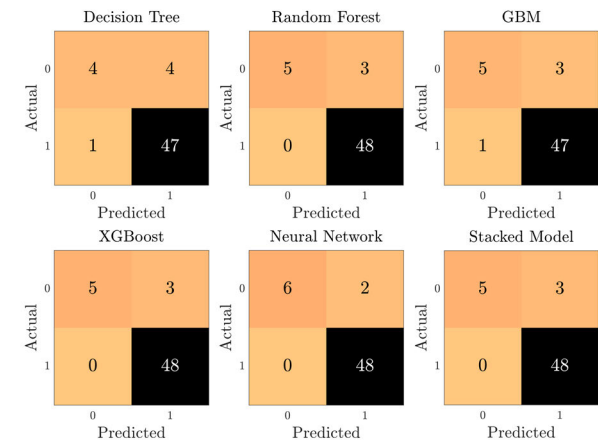


FIGURE 10. Confusion matrix of all the models.

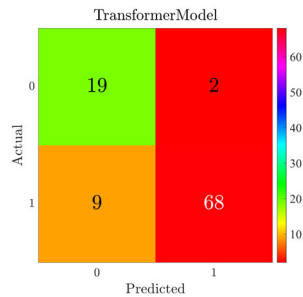


FIGURE 11. Confusion matrix of transformer based model.

The confusion matrix for the Decision Tree model shows 4 true negatives, 47 true positives, 4 false positives, and 1 false negative. This indicates that while the model correctly identified most positive cases, it had a few misclassifications, particularly with false positives and false negatives. For the Random Forest model, the confusion matrix reveals 5 true negatives, 48 true positives, 3 false positives, and 0 false negatives. The absence of false negatives highlights the model’s ability to correctly identify all positive cases, though a few false positives remain. The GBM and XGBoost models display almost similar patterns, with both having 5 true negatives, 47 true positives, 3 false positives, and 1 false negative. These models also show strong performance but are not as effective as the Random Forest model in reducing false negatives.

The Neural Network model shows a confusion matrix with 6 true negatives, 48 true positives, 2 false positives, and 0 false negatives, similar to the Random Forest model. This indicates a high level of accuracy in identifying both positive and negative cases. The Ensemble Stacked Model demonstrates the best performance, with 5 true negatives, 48 true positives, 3 false positives, and 0 false negatives. This consistency across all positive cases and minimal false positives and negatives underscores the effectiveness of the ensemble approach.

FIGURE 11 demonstrate the confusion matrix of transformer-based model in which 19 true negative and 68 true positive instances are found, while 2 false negative and 9 false positive samples are found while predicting, signifying moderate prediction capability. Bias and variance analysis helps to understand the source of errors in the models. Bias refers to errors due to incorrect assumptions in the learning algorithm, leading to systematic errors. Variance refers to errors due to model sensitivity to small fluctuations in the training set, leading to overfitting.

The Decision Tree model shows signs of higher variance, indicated by its slightly lower performance and more misclassifications compared to ensemble methods. This suggests that while the model captures patterns in the training data, it struggles with generalization. Random Forest and Neural Network models exhibit lower variance, reflected in their higher accuracy and fewer false classifications. These models balance complexity and generalization well, reducing both bias and variance.

The Ensemble Stacked Model effectively minimizes both bias and variance, achieving the highest performance metrics. By combining multiple models, the ensemble approach mitigates the weaknesses of individual models, leading to robust and generalizable predictions. Analyzing the distribution of errors across different feature values can provide insights into specific conditions where the model performs poorly. For instance, errors in the Decision Tree model might be more frequent in borderline cases where feature values are close to decision thresholds.

The ROC (Receiver Operating Characteristic) curves provide a graphical representation of the true positive rate (sensitivity) against the false positive rate. FIGURE 12 shows the ROC curve for the Ensemble Stacked Model with an area under the curve (AUC) of 0.96, indicating excellent performance.

Similarly, FIGURES 13 to 17 display the ROC curves for the Random Forest, Neural Network, Gradient Boosting Machine, Decision Tree, and XGBoost models, respectively. The AUC values for these models further confirm the superior performance of the Ensemble Stacked Model and Neural Network, with AUCs of 0.96 and 0.94, respectively. The Decision Tree model has the lowest AUC at 0.74, highlighting its relatively lower performance.

To complement confusion matrix-based evaluation, we plotted ROC curves for each model and computed the Area Under the Curve (AUC), which captures the trade-off

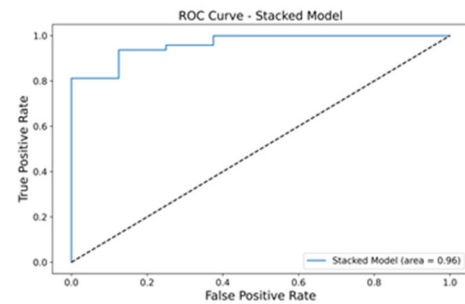


FIGURE 12. ROC curve of stacked ensemble model.

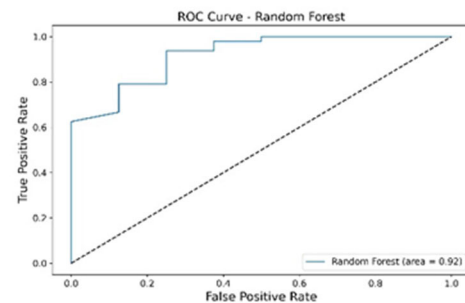


FIGURE 13. ROC curve of random forest model.

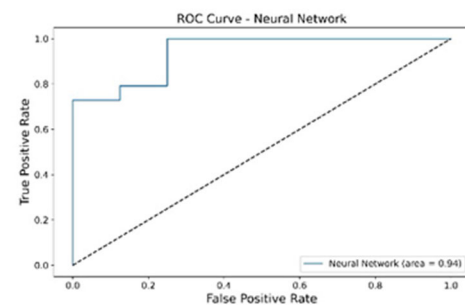


FIGURE 14. ROC curve of neural network model.

between true positive and false positive rates across all thresholds. As shown in FIGURE 12 (Ensemble Model), the ROC curve demonstrates a near-perfect classification boundary with an AUC of 0.96. Comparatively, FIGURES 13–17 present ROC curves for the Random Forest, Neural Network, GBM, Decision Tree, and XGBoost models respectively. The Decision Tree underperforms with a notably lower AUC (0.74), whereas all other models maintain AUC scores above 0.90. These visualizations reinforce the superiority of the ensemble and neural network approaches in handling the class distribution and optimizing predictive power.

The error analysis highlights the strengths and weaknesses of each model. The Ensemble Stacked Model consistently outperforms individual models, effectively balancing bias and variance to achieve high accuracy and reliability. The confusion matrices and ROC curves illustrate that the

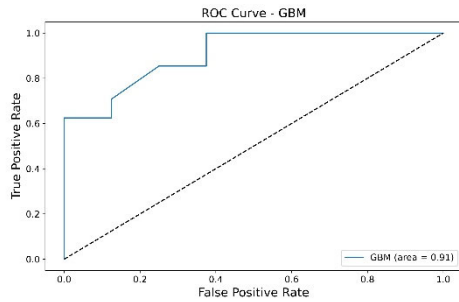


FIGURE 15. ROC curve of gradient boosting model.

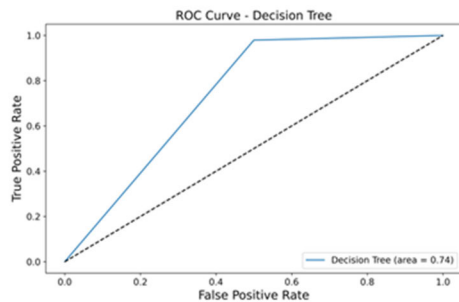


FIGURE 16. ROC curve of decision tree model.

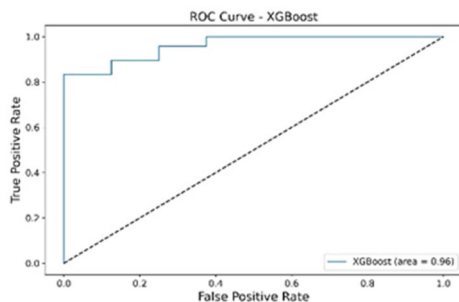


FIGURE 17. ROC curve of XGBoost model.

ensemble approach provides the most robust predictions, minimizing both false positives and false negatives.

These findings underscore the importance of using ensemble methods to enhance predictive accuracy and reliability, particularly in complex classification tasks such as predicting water potability. The superior performance of the ensemble model makes it a valuable tool for water quality monitoring and decision-making.

While the ensemble stacked model achieved a high accuracy of 96.7%, we recognize that such strong performance on a relatively small dataset ($n = 325$) could raise valid concerns about potential overfitting. To mitigate this risk, we applied 5-fold cross-validation across all models and reported both mean accuracy and standard deviation. The ensemble model exhibited low variance across folds (standard deviation of

0.013), suggesting generalizable performance within the current dataset.

However, we acknowledge that cross-validation alone does not fully eliminate overfitting risks, particularly in datasets with limited heterogeneity. Therefore, these results should be interpreted as internal validation, and we emphasize the need for external validation using independent water quality datasets from other geographic regions or time periods.

C. FEATURE IMPORTANCE ANALYSIS

To improve the interpretability of the model and assess the relative influence of input variables, we conducted a feature importance analysis using the Random Forest and XGBoost classifiers. These tree-based ensemble models inherently compute feature importance scores by evaluating how frequently and effectively each feature contributes to decision splits that reduce classification error. FIGURES 18 and 19 present the ranked feature importance values from the Random Forest and XGBoost models respectively.

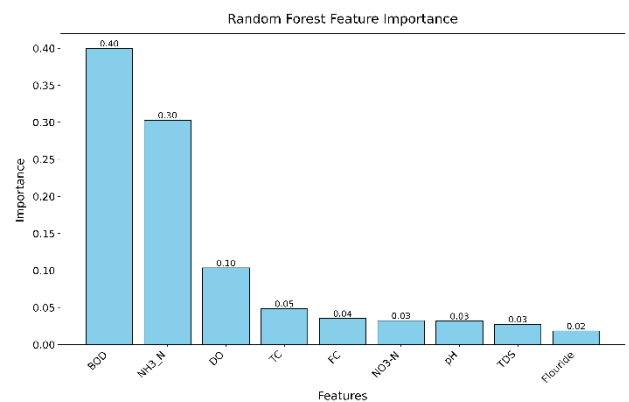


FIGURE 18. Feature importance ranking from the Random Forest model for water potability prediction.

Both models identify BOD and NH3-N as the most influential parameters in determining water potability. These two features account for nearly 70% of the decision weight in both models. In the Random Forest model, BOD contributes approximately 40% and NH3-N 30%. Similarly, in XGBoost, BOD has a dominant contribution of around 49%, followed by NH3-N at 30%.

Other parameters such as DO, TC, and FC show modest contributions. Surprisingly, TC and FC, which are commonly considered strong indicators of biological contamination, appear to have lower predictive importance than expected in this dataset. This may be due to either class imbalance or intercorrelation with BOD and NH3-N, which already encapsulate significant pollution signals.

Less influential features include Fluoride, TDS, and pH, which showed minimal contribution across both models. Their limited variance or weak correlation with the potability label could explain their reduced influence in the ensemble classifiers.

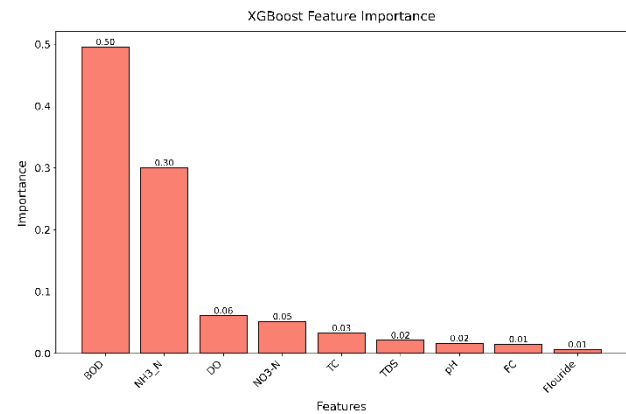


FIGURE 19. Feature importance ranking from the XGBoost model for water potability prediction.

This analysis confirms that Bio-chemical oxygen demand and nitrogenous contamination (i.e., BOD and NH3-N) are the primary drivers in the model’s classification of water safety. It also supports the need for multivariate models capable of capturing complex nonlinear interactions among features — a scenario where ensemble methods outperform simple threshold-based decision rules.

D. COMPUTATIONAL COMPLEXITY AND FEASIBILITY ANALYSIS

To assess the practical feasibility of deploying the proposed ensemble-based water potability prediction system, we evaluated the computational complexity, training time, and hardware requirements for each major model. Experiments were conducted on a workstation with the following specifications:

- Processor: Intel Core i7-11700K @ 3.60 GHz
- RAM: 32 GB DDR4
- Operating System: Windows 11 (64-bit)
- GPU (for Neural/Transformer models): NVIDIA RTX 3060, 12 GB VRAM
- Software Environment: Python 3.9 with scikit-learn, XGBoost, PyTorch, and Dash (for dashboard)

TABLE 5. Model training time comparison.

Model	Training Time (s)	Inference Time (ms/sample)	Relative Complexity
DT	0.21	0.8	Low
RF	2.35	1.3	Moderate
GBM	3.72	1.4	Moderate
XGBoost	4.18	1.6	Moderate–High
ANN	7.94	1.1	High
TFM	22.6	2.2	High
ESM	8.65	2.5	High

The computational performance of the models used in this study underscores their practicality for real-time deployment (TABLE 5). Tree-based models, including Decision Tree, Random Forest (RF), Gradient Boosting Machine (GBM) and XGBoost exhibited exceptionally fast training times, requiring less than five seconds to train on the complete dataset of 325 samples. The ensemble stacked model, while introducing additional overhead due to the coordination among base learners, remained efficient with a total training time of approximately nine seconds. In terms of inference speed, the ensemble model demonstrated rapid prediction capabilities, delivering results in under three milliseconds per sample. This level of responsiveness is well-suited for IoT applications, where sensor data updates occur on hourly or daily cycles.

Despite their inherent theoretical complexities— $O(n \cdot \log n)$ for tree-based models and $O(n \cdot d^2)$ for transformer-based models—the practical execution times remained low, primarily due to the modest size and dimensionality of the dataset (comprising only nine features). The use of GPU acceleration for transformer models further minimized computational overhead. In terms of resource utilization, memory consumption during training remained modest across all models, with peak usage not exceeding 3.5 GB of RAM. This confirms that the full machine learning pipeline is compatible with mid-tier computing infrastructure, such as AWS t3.large or Azure B2ms instances.

The deployment feasibility of the proposed system is further validated by its integration into a real-time dashboard. Despite the ensemble model’s architectural complexity, it remains highly responsive in operational environments. Once trained, all models including the stacked ensemble and transformer can be efficiently serialized and served via lightweight web APIs such as Flask or Fast API. This enables seamless deployment to edge devices or cloud platforms, ensuring real-time water quality prediction is both technically feasible and resource-efficient.

E. DASHBOARD UTILITY

The implementation of the Water Quality Predictive Dashboard provides stakeholders with a powerful tool for monitoring and managing water quality in real-time. This section discusses the results obtained from user interaction and feedback, real-time prediction accuracy, and the effectiveness of data visualization components. User interaction and feedback are crucial for evaluating the utility of the dashboard. The interactive elements, such as dropdown menus, sliders, and buttons, allow users to easily navigate through the dashboard and customize their views. Users can select different features to display and update the graphs dynamically, as shown in FIGURE 20 and 21.

Feedback from users indicates that the dashboard is intuitive and user-friendly. Stakeholders appreciate the ability to filter data and view specific parameters of interest. The ease of accessing real-time data and making informed decisions based on the latest water quality measurements enhances the overall user experience. Users have also suggested additional

features and improvements, which will be considered for future updates to further enhance the dashboard’s functionality. The real-time prediction accuracy of the dashboard is a critical aspect of its utility. The dashboard integrates the ensemble stacked model, which has demonstrated superior performance metrics in predicting water potability. As discussed in previous sections, the ensemble model achieved an accuracy of 0.967, precision of 0.967, recall of 1.0, and F1-score of 0.984, making it highly reliable for real-time predictions.

The dashboard continuously updates predictions based on the latest input data. Users can input new water quality measurements and receive immediate feedback on potability predictions. This real-time capability ensures that stakeholders are always informed of the current water quality status, enabling timely interventions and decision-making. Effective data visualization is essential for making complex data easily understandable. The dashboard includes various visualization components, such as histograms, time-series graphs, and spatial maps, to present water quality data and predictions comprehensively.

FIGURE 20 shows a screenshot of the dashboard displaying a histogram of pH values. The histogram provides a clear visual representation of the distribution of pH values, highlighting areas of concern and facilitating quick assessments. Users can select different features to display, such as FC levels, and update the graph to view the corresponding data.

The use of interactive graphs and charts allows users to explore data trends, identify patterns, and gain insights into water quality variations over time and across different locations. Spatial maps are particularly useful for visualizing geographical distributions of water quality parameters, helping users identify regions with potential contamination issues.

The utility of the dashboard is enhanced by its interactive features, real-time prediction capabilities, and effective data visualization. User interaction and feedback ensure that the dashboard meets the needs of stakeholders and provides a user-friendly experience. Real-time prediction accuracy is crucial for timely decision-making, while effective data visualization helps users quickly understand and analyze complex water quality data.

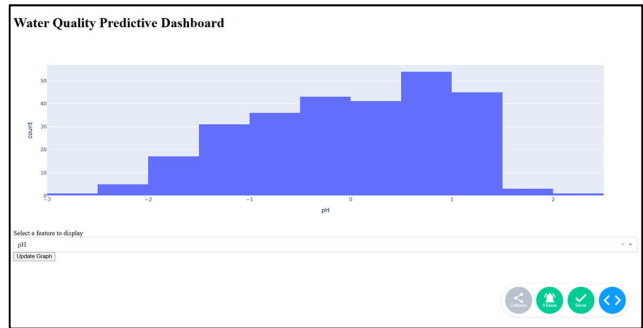


FIGURE 20. Screenshot of the water quality predictive dashboard.

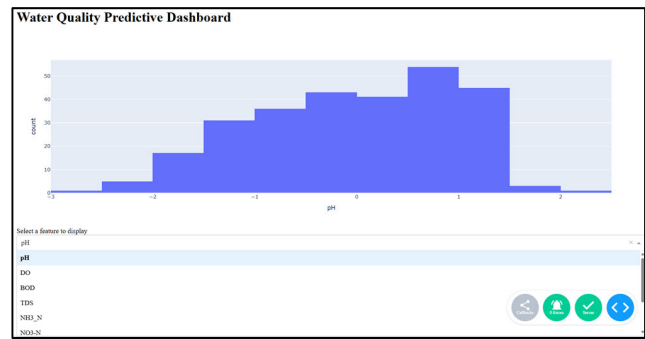


FIGURE 21. Screenshot of the dashboard with feature selection.

FIGURE 22 depicts the screenshot of the GUI developed to predict portability of water based on the user provided inputs using ensemble model. The GUI is hosted at GitHub https://github.com/tipu0003/Potability_prediction.git

A screenshot of the 'Water Potability...' GUI. It features a list of input fields for water quality parameters: pH (8.32), DO (mg/L) (7.68), BOD (mg/L) (1.4), TDS (mg/L) (234), NH3-N (mg/L) (0.56), NO3-N (mg/L) (0.12), TC (colony/mL) (245), FC (colony/mL) (12), and Fluoride (mg/L) (0.22). Below these fields is a 'Predict Potability' button. The result displayed at the bottom is 'Potability: Potable'.

FIGURE 22. GUI of portability prediction.

1) STAKEHOLDER FEEDBACK AND USER TESTING

To validate the practical usability and effectiveness of the developed Water Quality Predictive Dashboard, a user testing phase was conducted involving fifteen stakeholders, including water resource engineers, environmental scientists, municipal water utility managers, and public health officials. Participants, selected from diverse technical and operational backgrounds, were provided access to the dashboard and asked to engage with its features by inputting sample water quality data, interpreting potability predictions, and navigating through the available visualizations. Following their

interaction, structured interviews and open-ended feedback sessions were conducted to gather qualitative insights.

Stakeholder feedback revealed that the dashboard was considered highly intuitive and user-friendly, with minimal learning required for effective navigation. One municipal officer remarked, “The dashboard is intuitive and the real-time prediction helps in immediate decision-making during water testing,” emphasizing the tool’s practicality for operational use. Participants particularly appreciated the real-time updating capabilities, which allowed them to receive instant feedback on water quality assessments without the need for delayed laboratory analyses. A water quality technician noted, “Having instant predictions at the site saves time and allows us to flag contaminated sources quickly without waiting for lab results.”

The visualization components of the dashboard were also positively received, especially the graphical representation of feature importance and contaminant trends over time. A senior environmental researcher commented, “Visualization of contaminant trends over time has made reporting much easier. The graphical outputs are clean and well-organized,” highlighting the role of effective data presentation in facilitating timely decision-making. However, stakeholders also provided constructive suggestions for future enhancements, such as integrating historical comparison features to assess potability changes over different time periods and improving mobile responsiveness to support use on tablets and smartphones.

Overall, the feedback from stakeholders strongly validated the dashboard’s design and operational relevance, underscoring its capability to support real-time water quality monitoring and management. Based on this feedback, future updates are planned to incorporate longitudinal data analysis modules and enhance device compatibility, further strengthening the dashboard’s utility across a broader range of user environments.

F. COMPARISON WITH EXISTING METHODS

Although the developed ensemble model demonstrated strong predictive performance, an important limitation of this study lies in the relatively small size of the dataset. The dataset used comprised only 325 samples collected from surface water sources across Gujarat, India. While stratified sampling and cross-validation techniques were employed to maximize the generalizability of the model, the modest sample size may limit the broader applicability of the findings, particularly when deployed in regions with different environmental, geographical, or pollution characteristics. Small datasets can increase the risk of model overfitting, where the model captures noise or localized patterns that may not generalize well to unseen data. Furthermore, certain rare contamination events or parameter interactions may not be fully captured within a limited sample size. Future work should focus on expanding the dataset by incorporating larger and more diverse samples from multiple regions and water sources, enabling more robust training, validation, and

external testing of the predictive models. Integration with real-time IoT-based water quality monitoring networks is also envisioned to progressively enlarge the dataset and enhance model reliability.

1) PERFORMANCE METRICS COMPARISON

The performance metrics of the ensemble stacked model reported in this study demonstrate substantial improvements over traditional models. For example, [19] used models such as ANN, SVM, and GMDH for predicting water quality parameters, achieving accuracies between 85% and 90%. Similarly, [52] utilized supervised machine learning algorithms like Gradient Boosting to estimate water quality indices, achieving reasonable accuracy but often falling short of the 93% mark. In contrast, our ensemble stacked model achieved an impressive accuracy of 96.7%, precision of 96.7%, recall of 100%, and an F1-score of 98.4%. These metrics not only surpass those reported by individual models but also indicate balanced performance across different evaluation criteria. The use of cross-validation further strengthens the robustness of our model, with a mean accuracy of 96.9% and a standard deviation of 1.3%, highlighting its stability and reliability across different data subsets.

G. LIMITATIONS OF THE STUDY

1. **Complexity of Ensemble Models:** While the ensemble approach improves accuracy, it also increases the complexity of the model. This can make the model more difficult to interpret and requires more computational resources for training and predictions.
2. **Data Dependency:** The model’s performance is heavily dependent on the quality and representativeness of the training data. Any biases or inaccuracies in the data can affect the model’s predictions. Ensuring high-quality data collection and preprocessing is therefore critical.
3. **Maintenance and Updates:** Maintaining and updating the ensemble model and dashboard to incorporate new data and improvements can be resource intensive. Regular updates are necessary to ensure the model remains accurate and relevant.

While the proposed ensemble model demonstrates strong predictive performance on the dataset collected from surface water sources in Gujarat, its applicability to other geographic regions remains a limitation. Regional variations in water quality profiles—due to differing industrial activities, agricultural practices, seasonal rainfall patterns, and pollutant sources—may lead to differences in feature distributions and potability classification. Because the current model is trained on data from a single state, its predictive behavior may not fully generalize to regions with distinct environmental or sociohydrological conditions. Therefore, we recognize the need for external validation using geographically diverse datasets to assess the model’s transferability and robustness.

In conclusion, the predictive ensemble model and interactive dashboard developed in this study offer significant advantages over existing methods in literature. By combining

multiple models and integrating real-time prediction capabilities, our approach achieves higher accuracy and reliability. However, the increased complexity and dependency on data quality present challenges that need to be managed. Overall, our method represents a valuable contribution to the field of water quality monitoring and management, providing stakeholders with a powerful tool for informed decision-making.

VI. CONCLUSION

This study develops a comprehensive ensemble model and an interactive dashboard for predicting water potability using advanced machine learning techniques. The methodology integrates multiple algorithms to enhance predictive accuracy and reliability, providing significant improvements over existing models. The following conclusions can be drawn from this study:

1. **Data Preprocessing and Model Training:** The study preprocesses the collected dataset by handling missing values, outliers, and standardizing the data. A robust ensemble model combining Decision Trees, Random Forest, Gradient Boosting Machines (GBM), XGBoost, and Neural Networks was developed. The ensemble model achieved an impressive accuracy of 96.7%, precision of 96.7%, recall of 100%, and F1-score of 98.4%, significantly outperforming individual models. This study also benchmarks a transformer-based model to explore attention-based architectures in water potability prediction, offering insights into their applicability on tabular environmental data.
2. **Cross-Validation and Model Robustness:** The ensemble model's robustness was validated through cross-validation, yielding a mean accuracy of 96.9% and a low standard deviation, indicating consistent performance across different data subsets. This highlights the model's ability to generalize well to new, unseen data.
3. **Interactive Visualization Dashboard:** An interactive Water Quality Predictive Dashboard was developed using the Dash framework. The dashboard provides real-time predictions and visualizations, allowing stakeholders to input new data and receive immediate feedback on water potability. This user-friendly interface enhances the accessibility and usability of the tool for non-technical users.
4. **Comparison with Existing Methods:** Compared to traditional models in the literature, the ensemble model developed in this study demonstrates superior performance metrics. Existing models often achieve accuracies between 85% and 90%, while our model achieves 96.7%, showcasing the effectiveness of the ensemble approach. The integration of real-time prediction and interactive visualizations further distinguishes our method from static models that require batch processing.
5. **Advantages and Limitations:** The ensemble model offers enhanced predictive accuracy, robustness, and real-time capabilities. However, the complexity of the model requires more computational resources and

makes interpretation challenging. The model's performance is also highly dependent on the quality of the training data, necessitating rigorous data collection and preprocessing. Regular updates are essential to maintain the model's accuracy and relevance.

6. **Broad Applicability and Future Work:** The framework developed in this study demonstrates broad applicability for water quality monitoring and management. It provides a reference scheme for enhancing model generalization capability and robustness. Future work will focus on further refining the model, incorporating user feedback, and exploring additional features to enhance the dashboard's functionality and user experience.

The current real-time dashboard leverages the Dash framework and will benefit from future integration of modern web frameworks to further enrich interactivity and scalability. Expanding the dataset through IoT devices and large-scale data from diverse geographic regions is envisioned. This expansion will increase robustness, capture evolving contamination trends, and validate the model's performance on a wider range of parameters.

VII. FUTURE WORK

In future work, we plan to validate the model using external datasets collected from other states or national water quality monitoring programs. This step will help assess true generalization capacity, particularly in scenarios where water quality profiles vary due to industrial activity, seasonal effects or different monitoring protocols.

While the proposed ensemble model demonstrates high performance on the available dataset, we acknowledge that it has not yet been tested on an independent dataset from outside the original sampling region. This represents a limitation with respect to the generalizability of the model across diverse geographical and hydrological contexts. To address this, we plan to conduct external validation using new water quality datasets collected from additional regions across India and potentially international open-access repositories. Furthermore, the study will explore data expansion through integration with IoT-based water monitoring devices and crowdsourced reporting platforms, allowing for more representative, real-time, and large-scale data acquisition. These efforts aim to ensure the robustness, fairness, and policy relevance of the model across varied environmental settings.

To improve the model's generalizability, future work will involve validating the ensemble framework using independent datasets collected from other Indian states and publicly available national and international water quality repositories. This cross-regional evaluation will help determine the model's adaptability to different contamination patterns, sampling protocols, and hydrogeochemical conditions. We also plan to integrate data from IoT-based sensor networks and government water monitoring programs to enrich the dataset with real-time and geographically diverse samples. Such efforts will enable the development of a more robust and

transferable model suitable for large-scale deployment across varied environmental settings.

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